A 2nm 234.4 TOPS/W and 511.9 TOPS/mm^2 Digital Computing-in-Memory Compiler with Multiple MAC Units per Weight and Multiple Data Format Support

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Abstract

This paper presents a Compute-in-memory (CIM) compiler featuring multiple data format and multiple MAC units per weight, supporting flexible data format, better utilization and throughput. We implemented various configurations in a test chip with our 2nm nanosheet technology node and test chip measurement results show V_{MIN} < 0.38V, 234.4 TOPS/W at 0.5V and 511.9 TOPS/mm^2 at 1.2V.

Introduction

CIM is being widely explored to minimize power consumption related to data movement and multiply-and-accumulate (MAC) operations for AI edge devices. Compared to analog based CIM, digital based CIM (DCIM), which includes small, distributed SRAM banks and a customized MAC unit, realizes massive-parallel computation with no accuracy loss and a better power-performance-area (PPA) scaling with the advanced technology. New applications demand higher area (TOPs/mm^2) and energy efficiency (TOPs/W) across a diverse set of data formats and general matrix multiplication (GeMM) shapes (Fig. 1) [1-3]. To meet these demands, this paper introduces a 2nm [4] SRAM-based CIM compiler that supports multiple data formats and multiple MAC units per weight cell. First, with support for several integer data formats at minimal overhead, DCIM can support mixed-precision quantization for most modern neural networks. Next, DCIM compiler supports multiple MAC units per weight cell to exploit weight reuse and optimize both TOPs/mm^2 and GeMM utilization. We implemented the DCIM macros with CIM BIST and ATPG in a test chip and confirmed ultra-low voltage MAC operation down to 380 mV. The CIM achieves 88.8 TOPS/W and 174.2 TOPS/mm2 with INT8 format and 234.4 TOPS/W and 511.9 TOPS/mm2 with INT4 format.

2nm CIM Compilers with INT Format

Fig. 2 summarizes the overview of 2nm CIM compilers with INT format. Compiler-A contains one MAC unit per weight, while compiler-B features two. In the compiler-B, weight data from bit cell array is shared among two MAC units. On the other hand, each MAC unit receives independent input activation (XINA and XINB) from outside of CIM macro and outputs the MAC are calculated separately (NOUTA and NOUTB). Both input activation and weight in a MAC unit supports data format with signed/unsigned INT8 and INT4. Furthermore, when both input activation and weight data formats are INT4, the MAC size (i.e. the number of input channels) becomes double (e.g., one macro supports 16 input channels with INT8xINT8 and 32 input channels with INT4xINT4). Compiler-B can also extend input activation bit width to 16bit by appending NOUTA to NOUTB after an 8-bit shift-up although this halves the number of MAC units. The method for supporting flexible input channel size, output channel size and the number of rows in each bit cell array is similar to that described in [3]. In leaf cell tiling, the input channel size, output channel size and the number of rows can be similar to the number of banks, word width and word depth in SRAM compiler design. Configuration range in our 2nm INT CIM compilers become 16 to 64 input channels with step = 16 when supporting INT8 format, 2 to 8 output channels with step = 1, and 2 to 18 SRAM storage rows with step =2.

Fig. 3 shows the block diagram of MAC unit in the compiler-B. Multipliers are divided into four groups (MUL0 to MUL3), with each multiplier handling 4b x 4b. For INT8 input x INT 8 weight MAC operations, all multipliers are active. Conversely, MUL1 and MUL2 are gated and output zero when supporting INT4 input x INT4 weight MAC operations. The MAC unit also supports INT8 input x INT4 weight and INT4 input x INT8 weight cases by disabling MUL0 and MUL1 and MUL0 and MUL2, respectively, although MAC size cannot exceed the MAC size in INT8 x INT8 case. There are two adder trees (adder tree1 on the MSB side and adder tree0 on the LSB side) within the MAC unit to efficiently support double MAC size with INT4 input x INT4 weight. Because MUL1 and MUL2 output zero in INT4 input x INT4 weight mode, no carry propagation from MUL0 and MUL3 occurs in multiplier before adder tree for accumulation. In the adder tree1, 8bit shift happens when we support INT8 format. On the other hand, in the INT4 input x INT4 weight mode, partial sums from the adder tree1 and adder tree2 are directly accumulated without shift operation. In order to support INT16 input x INT8 weight, NOUTB and NOUTA after shifting 8-bit can be appended.

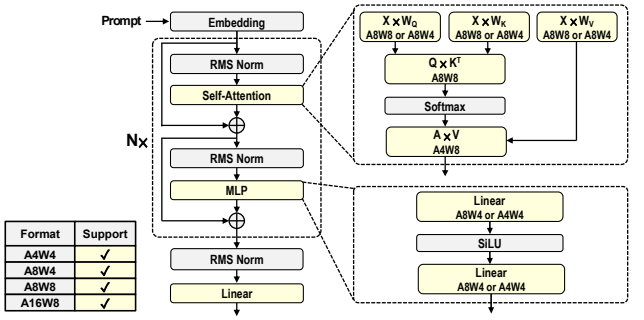
Fig. 4 depicts the tiling floorplan with MAC unit and bit cell array leaf cells, which are crucial components in the CIM macro in compiler-B. One basic MAC unit leaf cell supports MAC operation with eight inputs and one output. Two look-up-table (LUT) based MAC units are sandwiched between two bitcell arrays. In contrast, compiler-A has only one MAC unit. As previously explained, the number of cells and wordlines (WLs) changes according to the required number of rows in a bit cell array. We adopted a latch-based 12T bitcell [5] with a size of 0.0437 μm^2 in the 2nm technology. By placing four weights with eight bits and two rows with an interleaved WL scheme (resulting in 64 cells per one output channel in the WL direction), we can maintain global routing tracks for sum propagation, enabling the implementation of compiler-B with two MAC units. This approach also reduces dynamic power because of shorter global routing in the adder tree. Another benefit of this topology is higher write bandwidth, allowing to update four weights in a single cycle.

Fig. 5 compares TOPS/W and TOPS/mm^2 between compiler-A and -B at 0.5V. Focusing solely on MAC operation, the compiler-A realizes 8% better energy efficiency when the input toggle rate is 25% because global routing is shorter than in compiler-B. However, when considering row switch (RSW), energy difference between compiler-A and -B diminishes because the row switch energy (local SRAM read operation) can be shared with two MAC units in compiler-B. When a row switch happens every cycle, compiler-B's energy efficiency becomes 2% better than compiler-A's. Because of gating logic for power saving and shorter critical path when supporting INT4 format, TOPS/W and TOPS/mm^2 become more than 2x better than INT8 format.

Measurement Results

Fig. 6 shows a 2nm test chip micrograph. We implemented various configurations to verify the CIM compiler. The CIM compiler test unit includes not only CIM macros but also CIM BIST and ATPG in order to ensure test coverage. Figs. 7 and 8 summarize the minimum operating voltage (V_{MIN}) and maximum operating frequency (F_{MAX}) for the biggest configurations, respectively. We observed that V_{MIN} at -40 °C can be lower than 0.38V with 95% yield. When supporting INT8 format, F_{MAX} of compiler-A and -B at 1.2V and -40 ºC are 2.0 GHz and 1.7 GHz, respectively. Even at 0.5V, F_{MAX} is faster than 500 MHz. Fig. 9 is the power measurement results for compiler-3B. With an input toggle rate of 25% and 0.5V, TOPS/W ranges from 88.8 to 145.6 when supporting INT8. If the format is INT4, TOPS/W ranges from 234.4 to 324.4. Fig. 10 summarizes comparison with prior works.

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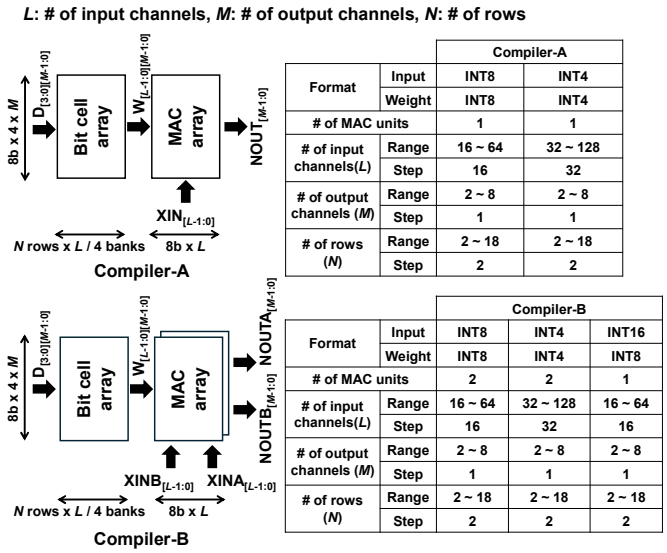


Input	Output	MAC	Macro	TOPS/mm² @0.5V		Utilization (%)	
				2rows	18rows	ResNet-18	MobileNet-v4
16	8	1	1	42.2	32.3	99.2	37.0
16	8	2	1	46.6	39.7	99.0	35.6
32	8	1	1	46.9	35.0	99.0	28.2
32	8	2	1	51.1	42.9	98.5	26.7
64	8	1	1	49.8	36.6	97.2	15.8
64	8	2	1	53.6	44.6	89.8	14.8
128	8	1	1	NA	NA	85.6	8.4

Similar utilization & better TOPS/mm²

Similar TOPS/mm² & better utilization

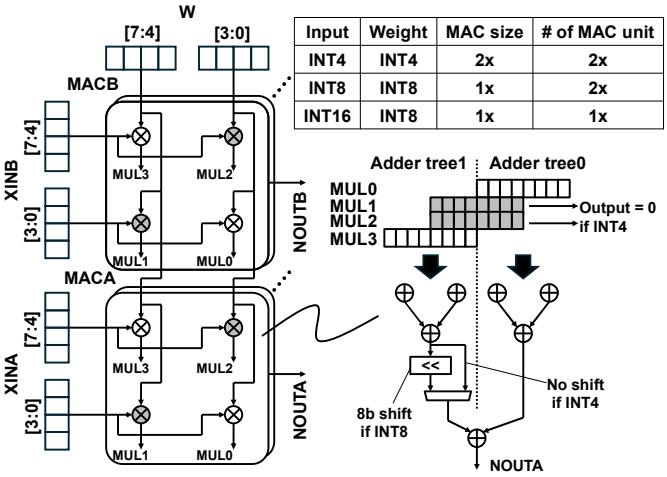
Fig. 1. Multiple data format and hardware utilization.



Format		Compiler-A	
	Input	INT8	INT4
	Weight	INT8	INT4
# of MAC units		1	1
# of input channels(*L*)	Range	16 ~ 64	32 ~ 128
	Step	16	32
# of output channels (*M*)	Range	2 ~ 8	2 ~ 8
	Step	1	1
# of rows (*N*)	Range	2 ~ 18	2 ~ 18
	Step	2	2

Format		Compiler-B		
	Input	INT8	INT4	INT16
	Weight	INT8	INT4	INT8
# of MAC units		2	2	1
# of input channels(*L*)	Range	16 ~ 64	32 ~ 128	16 ~ 64
	Step	16	32	16
# of output channels (*M*)	Range	2 ~ 8	2 ~ 8	2 ~ 8
	Step	1	1	1
# of rows (*N*)	Range	2 ~ 18	2 ~ 18	2 ~ 18
	Step	2	2	2

Fig. 2. 2nm CIM compilers and supporting range.



Input	Weight	MAC size	# of MAC unit
INT4	INT4	2x	2x
INT8	INT8	1x	2x
INT16	INT8	1x	1x

Fig. 3. MAC architecture supporting multiple data format.

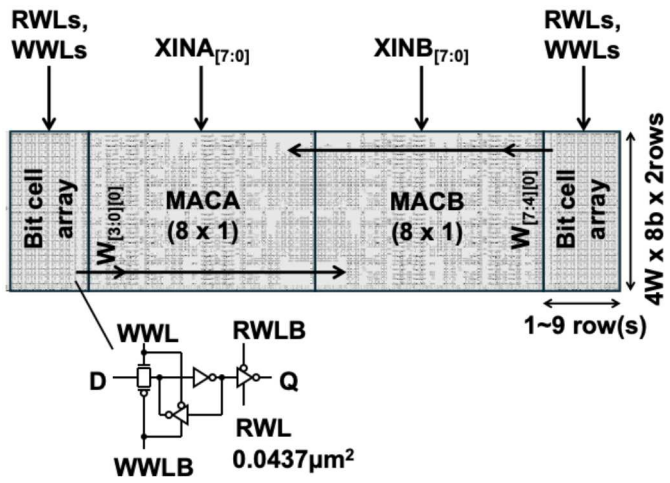


Fig. 4. Tiling floorplan with MAC unit and bitcell array leaf cells for one output channels.

		TOPS/mm² @0.5V		TOPS/W @0.5V		
		2rows	18rows	MAC	MAC + RSW/4cycle	MAC + RSW/cycle
Compiler-A	INT8xINT8	49.8	36.6	95.6	74.1	46.3
	INT4xINT4	110.2	81	250	193	121.6
Compiler-B	INT8xINT8	53.6	44.6	88.8	73.4	47.3
	INT4xINT4	119.1	99.1	234	191.9	124

Fig. 5. TOPS/W and TOPS/mm² with 64 input x 8 output configs.

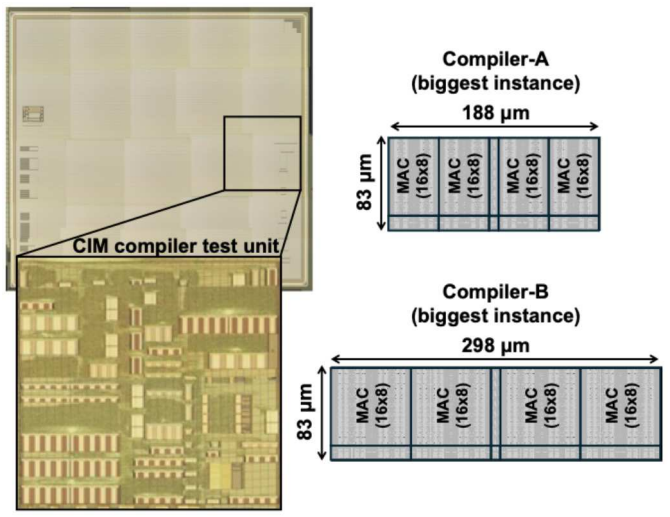


Fig. 6. Micrograph of test chip including CIM compiler test unit.

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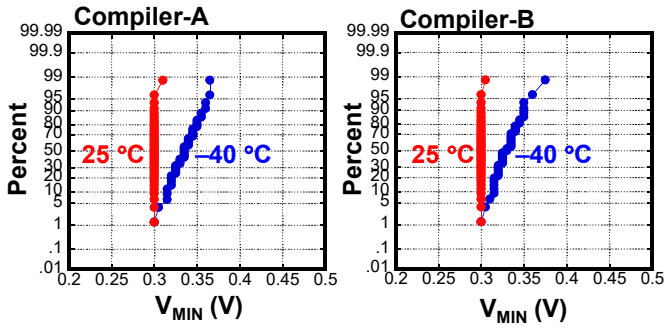


Fig. 7. V_{MIN} measurement results (VDD < 0.3V is not measured).

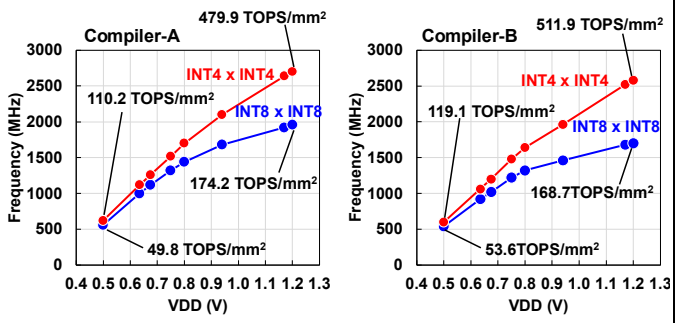


Fig. 8. F_{MAX} measurement results at –40 ºC

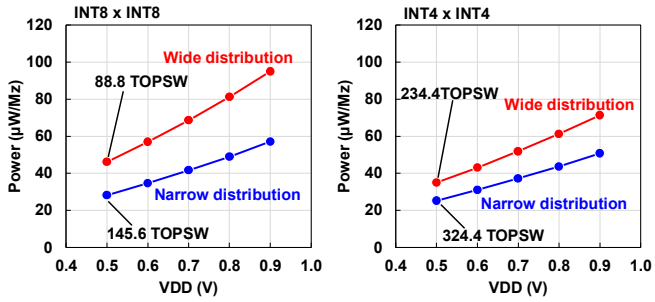


Fig. 9. Compiler-B power measurement results.

	ISSCC '23 [6]	ISSCC '24 [7]	VLSI '25 [3]	This work	
Technology	18nm	3nm	3nm	2nm	
Data format (Input / Weight)	INT8 / INT8	INT12 / INT12	INT8 / INT8	INT8 / INT8 INT4 / INT4	INT8 / INT8 INT4 / INT4 INT16 / INT8
Voltage	0.525-1.0V	0.36-1.1V	0.37-1.1V	0.38-1.2V	
Compiler support	No	No	Yes	Yes	
Input Ch	32	72	16 ~ 64	16 ~ 64 (INT8 / INT8) 32 ~ 128 (INT4 / INT4)	16 ~ 64 (INT8 / INT8 & INT16 / INT8) 32 ~ 128 (INT4 / INT4)
Output Ch	64	4	2 ~ 8	2 ~ 8	
Weight set	32	18	2 ~ 18	2 ~ 18	
MAC unit	1	1	1	1	2 (INT8 / INT8 & INT4 /INT4) 1 (INT16 / INT8)
TOPS/mm2	3.4 @1.0V	55.0@0.9V	90.2@1.1V	174.2 (INT8 / INT8) 479.9 (INT4 / INT4) @1.2V	168.7 (INT8 / INT8) 511.9 (INT4 / INT4) @1.2V
TOPS/W	14.5 @0.525V	32.5 @0.55V	70.4@0.5V	95.6 (INT8 / INT8) 250.4 (INT4 /INT4) with 18 weight set @0.5V 101.2 (INT8 / INT8) 264.9 (INT4 / INT4) with 2 weight set @0.5V	88.8 (INT8 / INT8) 234.4 (INT4 / INT4) with 18 weight set @0.5V 94.0 (INT8 / INT8) 248.0 (INT4 / INT4) with 2 weight set @0.5V

Fig. 10. Comparison table.

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